

AIML320/AIML420 — *Artificial Intelligence*

Assignment 3: Planning and Scheduling, and Neural Networking

15% of Final Mark — Due: 23:59 Wednesday 28 May 2026

IMPORTANT

All requirements specific to AIML420 students are marked as (**AIML420 mandatory**). AIML320 students may choose to complete the tasks marked as (AIML420 mandatory). Completing it correctly can earn up to 2 bonus marks. Bonus marks will only be added if the total mark (including bonus) does not exceed 100; any excess will be discarded.

Part 1: Job Shop Scheduling (40 marks)

Objectives

The goal of this assignment is to help you understand the basic concepts of planning and scheduling, and simple algorithms for them. In particular, the following topics should be reviewed:

- Investigate a job shop scheduling problem,
- Implement a heuristic to solve a scheduling problem.

Problem Description

You are required to find solutions (schedules) for a job shop scheduling problem, with 3 jobs and 2 machines, as shown in the table below.

Job	ArrivalTime	Operation	Machine	ProcTime
J_1	0	O_{11}	M_1	50
		O_{12}	M_2	25
J_2	10	O_{21}	M_2	30
		O_{22}	M_1	35
J_3	20	O_{31}	M_1	40
		O_{32}	M_2	20

- (Number of operations) Each job J_j has two operations O_{j1} and O_{j2} .
- (Precedence constraint) The operations must strictly follow the precedence constraint. That is, processing of O_{j2} ($j = 1, 2, 3$) cannot start before O_{j1} has been completely processed.
- (Arrival time) Each job has an arrival time (ArrivalTime). For each job J_j , processing of the first operation O_{j1} cannot start earlier than the job's arrival time.
- (Resource constraint) Each machine can process at most one operation at a time. Each operation can only be processed by a particular machine. For example, operation O_{11} can only be processed by machine M_1 .

Solution/Schedule Representation

A solution/schedule for a job shop scheduling problem is a sequence of actions. Each action is composed of the processed operation, the machine to process the operation, and the starting time. The finishing time of an action is the starting time plus the processing time of the processed operation. The actions are sorted in the ascending order of starting time, i.e., the former action starts no later than the latter one.

In this assignment, the following format is adopted to represent a schedule:

$$Process(O_{11}, M_1, 0) \rightarrow Process(O_{21}, M_2, 10) \rightarrow \dots,$$

where $Process(o, m, t)$ stands for an action that processes operation o with machine m starting at time t .

You do not need to have codes for questions in Part 1.

Questions

- (10 marks) Suppose you are given a schedule whose action sequence is as follows: $Process(O_{11}, M_1, t_1) \rightarrow Process(O_{21}, M_2, t_2) \rightarrow Process(O_{31}, M_1, t_3) \rightarrow Process(O_{12}, M_2, t_4) \rightarrow Process(O_{22}, M_1, t_5) \rightarrow Process(O_{32}, M_2, t_6)$. Since the sequence of actions is sorted in non-decreasing order of the starting time, you know that $t_1 \leq t_2 \leq t_3 \leq t_4 \leq t_5 \leq t_6$.

Calculate the **earliest starting time** for each of the six actions, i.e., t_1 to t_6 . You can draw a gantt chart to help you think.

Hint: the earliest starting time of an action is the later time between the earliest ready time of the operation and the earliest idle time of the machine.

- (10 marks for AIML320 and 5 marks for AIML420) For the solution given in Question 1, find the **completion time** of each job, which is the finishing time of its last operation. Then, calculate the **makespan** of the solution, which is defined as the maximum completion time of all the jobs.
- (10 marks for AIML320 and 5 marks for AIML420) Write the **final solution** obtained by the *Shortest Processing time (SPT)* dispatching rule. You may draw a figure (gantt chart) to help you find the solution.
- (5 marks) For the solution obtained by the SPT rule, calculate the completion time of each job and the makespan. Compare the makespan of this solution with that obtained in Question 1 to find out which solution is better in terms of makespan.

Note: the solution in Question 1 is obtained by the First-Come-First-Serve (FCFS) rule.

- (5 marks) The two compared solutions are obtained by the SPT and FCFS rules, respectively. If one solution is better than the other, does it mean that the rule that generates the better solution is better than the other rule? Why or why not?
- (AIML420 mandatory, 10 marks)** Often in practice, neither the SPT nor FCFS rules are good enough for solving the job shop scheduling problem. Suggest two methods to solve the job shop scheduling problem. One method should consider the problem to be static (all information is known in advance), and the other method should consider it to be dynamic (e.g. unpredicted job arrivals can happen in real time). **Clearly describe your methods (e.g. algorithm framework, solution representation/encoding, search operators).**

Part 2: Neural Networks (60 marks)

There are three tasks for this part of the assignment. (Plus an extra task for AIML420 students.)

Task a: Training a Simple Perceptron on Linearly Separable Data (20 marks for AIML320, 15 marks for AIML420)

Your first task is to implement a simple perceptron from scratch and evaluate how well it performs on a binary classification task using a predetermined train and test set.

You will use:

- `SeaSynTrain.csv` as the training set
- `SeaSynTest.csv` as the test set

Each dataset contains:

- **3 continuous features:** `attrib1`, `attrib2`, and `attrib3`
- **1 target column:** `class`, with values 0 or 1

Submit your implementation of the Perceptron, along with the code for evaluating its performance, in a single file called `perceptron`.

Instructions

- Use **accuracy** as your evaluation metric.
- Train the perceptron using different numbers of epochs:
[1, 5, 10, 15, 20, 50, 60, 80, 100, 120, 150, 200]
- In your report, show the test set accuracy after training for each number of epochs.
- Then, answer the following:

How does increasing the number of epochs affect the model's accuracy?

You can present your results in a table like this:

Model (epoch)	Accuracy
Perceptron (1)	
Perceptron (5)	
Perceptron (10)	
⋮	
Perceptron (150)	
Perceptron (200)	

Task b: Training a Simple Perceptron on Non-Linearly Separable Data (20 marks for AIML320, 15 marks for AIML420)

In this task, you will reuse your Perceptron implementation to evaluate its performance on a dataset that is **not linearly separable**. This will help you explore the limitations of a linear classifier. Precisely, you will use:

- `RingSynTrain.csv` as the training set
- `RingSynTest.csv` as the test set

Each dataset contains:

- **2 continuous features:** `x1` and `x2`
- **1 target column:** `class`, with values 0 or 1

Use the Perceptron implementation from Task (a) and evaluate (using the accuracy metric) on this dataset.

Instructions

- Use **accuracy** as your evaluation metric.
- Train the perceptron using different numbers of epochs:
[1, 5, 10, 15, 20, 50, 60, 80, 100, 120, 150, 200]
- In your report, show the test set accuracy after training for each number of epochs.
- Then, answer the following:

How well does the perceptron perform on this dataset?

You can present your results in a table like this:

Model (epoch)	Accuracy
Perceptron (1)	
Perceptron (5)	
Perceptron (10)	
⋮	
Perceptron (150)	
Perceptron (200)	

Task c: Training a MLP on Non-Linearly Separable Data (20 marks for AIML320, 15 marks for AIML420)

In this task, you will evaluate how a simple Multi-Layer Perceptron (MLP) handles the same non-linearly separable dataset from Task b. Unlike the perceptron you implemented from scratch, here you must use an existing implementation, for example, from `scikit-learn` or another library of your choice.

Use the same dataset as in Task b: i.e. `RingSynTrain` (for training) and `RingSynTest` (for testing).

Instructions

- Use a standard MLP classifier implementation from a library (e.g., `sklearn.neural_network.MLPClassifier`).
- Use a simple architecture — for example, one hidden layer with 4 to 10 neurons. Submit your code in a file called `mlp`.
- You don't need to test lots of configurations. Just pick one setup that seems reasonable, run it, and evaluate it using **accuracy**. Then in your report, answer the questions below:

How does the performance of the MLP compare to the Perceptron from Task b on this dataset?

Was the MLP able to better capture the structure of the data? Why or why not?

Task d: Activation Function Comparison (AIML420 mandatory, 15 marks)

In this task, you will investigate the effect of different activation functions on MLP performance.

Dataset:

`RingSynTrain.csv` (training set) and `RingSynTest.csv` (test set) Features: `x1`, `x2` Target: class (0 or 1)

Instructions:

1. Train MLPs with two different activation functions: `sigmoid` and `relu`.
2. Use the same architecture as in Task c (i.e. one hidden layer with 4–10 neurons). (You can use the same `mlp` code file, or submit a second one, `mlp2`.)
3. Evaluate each model on `RingSynTest.csv` and compare convergence speed and test accuracy.
4. In your report, answer the following questions:
 - Which activation function performed best and why?
 - How does activation choice affect learning dynamics?

Submission Guidelines

Submission Requirements

1. Please provide a README file that specifies how to run (and compile) your programs on the ECS School machines. If the markers can't run your code, you will lose marks. Add necessary comments in your program, and make sure to include ALL the required libraries and dependent files.
2. **readme.txt**: The **readme file** presents the detailed steps and commands on how to run your programs using the command line. If your programs cannot run properly, then you should provide the **list of bugs** in this file. On the other hand, if your program can run properly, then you should include the **sample terminal output** of your program in this file.
3. You should use **a single PDF** report for all parts, and label questions clearly. This document allows you to practice your technical writing skills. It should contain clear answers to all questions of this assignment. In general, **readers should be able to understand what you have done purely from the report, without looking at other sources of information (e.g., source code) provided by you.**

In summary, your submission should comprise one PDF report, one README.txt file, and several programs.

Submission Method

The PDF version of the assignment report and your programs and readme files should be submitted electronically through the Web submission system (accessible from the AIML320 or AIML420 course website) by the due date.

Late Penalties

The assignment must be submitted on time unless you have made a prior arrangement with the **course coordinator** or have a valid medical excuse. We use the ECS extension system for all extension requests. Please make a request there if you think you have a valid reason.

The penalty for assignments that are handed in late without prior arrangement is one grade reduction per day. Assignments that are more than one week late will not be marked.

Remember that you have three late days for this course (which can be used fractionally), but these apply across the whole course, not per-assignment. Please save some late days in case you need them later on.

Plagiarism

Plagiarism in programming (copying someone else's code) is just as serious as written plagiarism, and is treated accordingly. Make sure you explicitly write down where you got code from (and how much of it) if you use any other resources besides from the course material. Relying heavily on borrowed code may lead to deductions in marks, whereas plagiarism warrants a zero mark and may entail disciplinary actions.

Use of Generative AI

Don't use Generative AI to produce complete code solutions or report sections. That's not helpful for your learning. If we suspect you are relying too much on AI, we reserve the right to ask you questions prior to assigning marks.

- If you do use Gen AI to help with code or report, you should make sure you understand any content included. You should also rigorously test generated code, to ensure it works as expected.
- In your report, you should disclose any uses of Gen AI that you made.

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